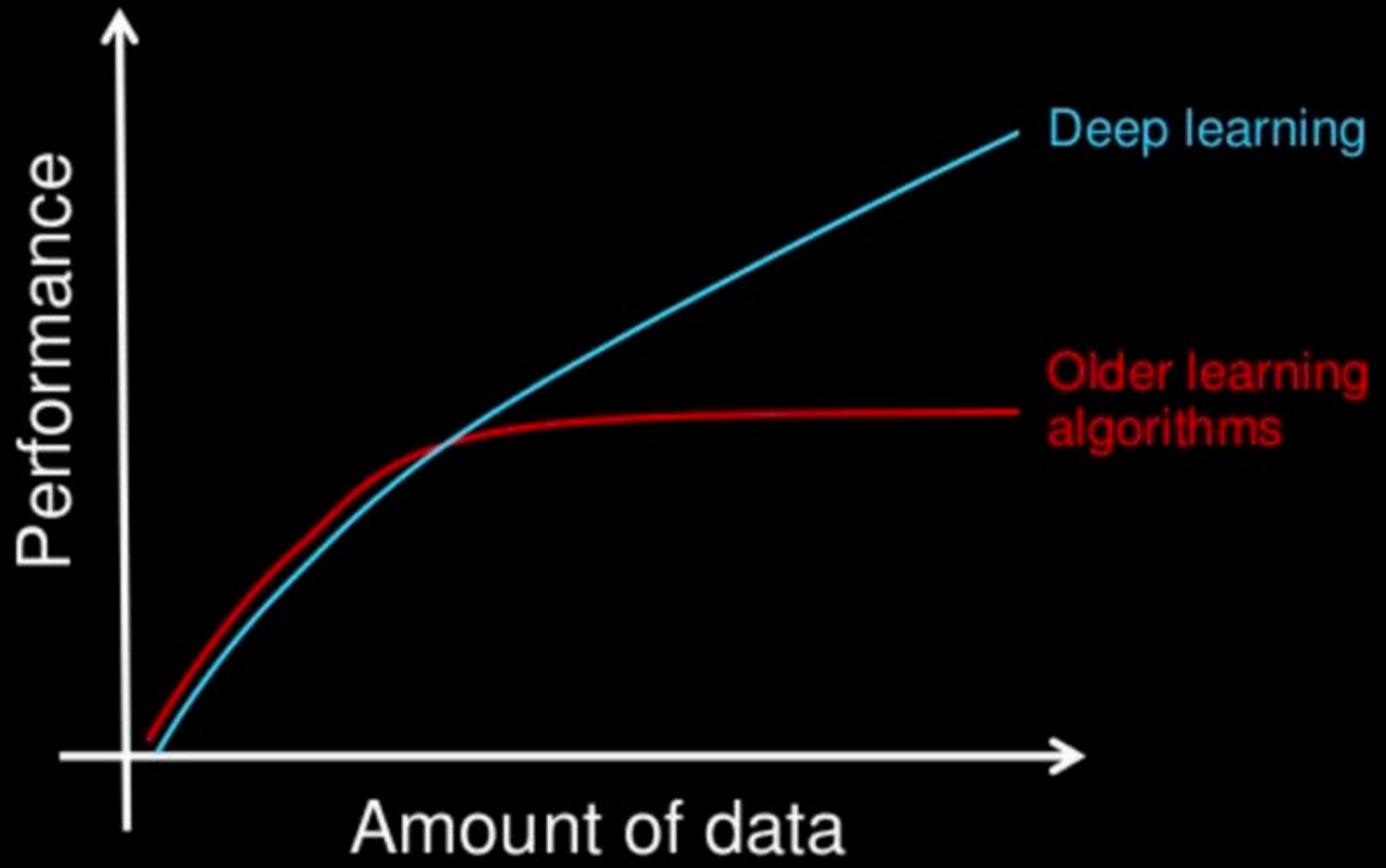
Why deep learning



How do data science techniques scale with amount of data?

[Slide credit: Andrew Ng]

Few-shot Learning



Which of these is an example of the same concept as the item in the box?

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[Lake, Salakhutdinov, Tenenbaum, 2015]

"Deep learning"



Human learning



Representations

Models

Skills

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How can we give deep nets prior knowledge?

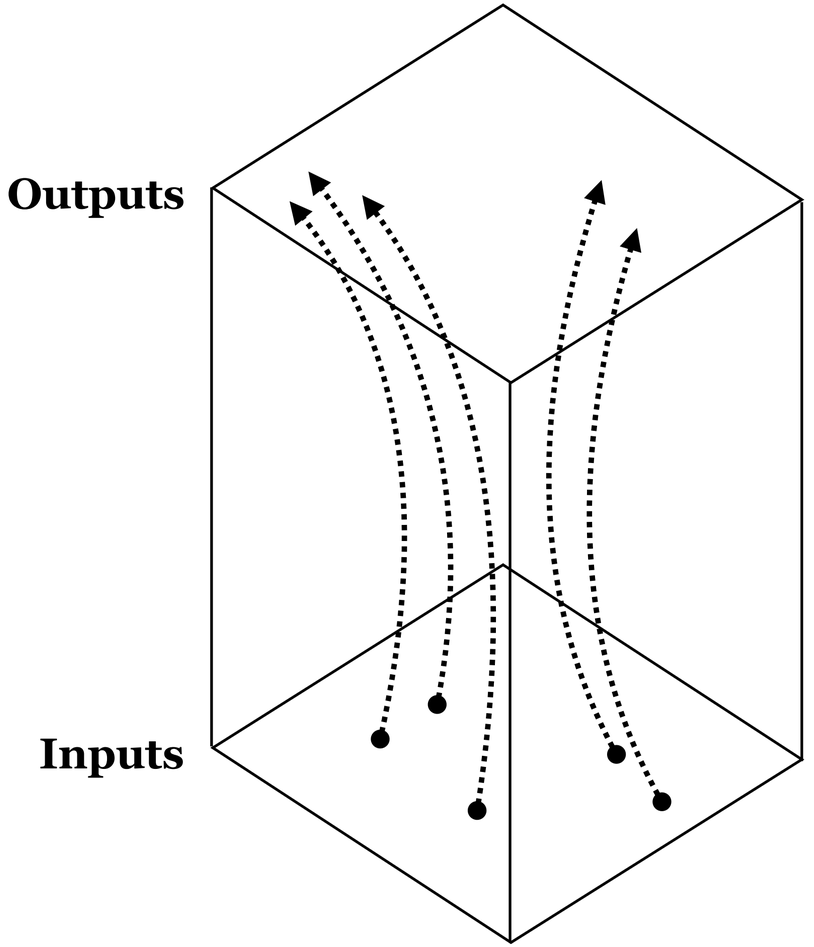
How can we give deep nets prior knowledge?

Just “pretrain” them on prior tasks!

Then transfer knowledge from the pretrained nets to solve novel tasks.

One view: the ***point*** of deep learning is to enable problem solving with little data

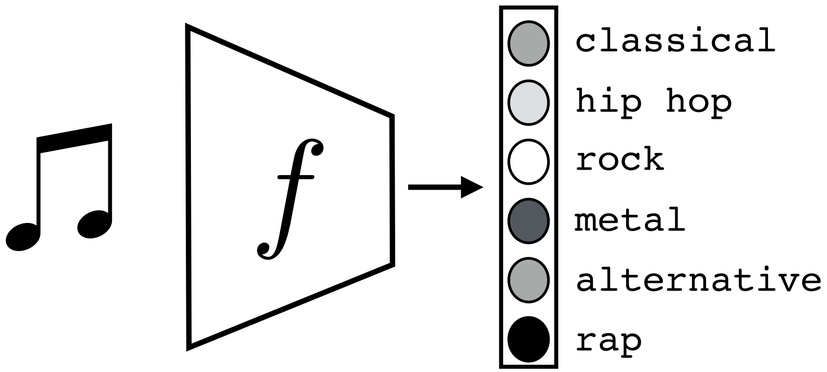
1. **Knowledge about the mapping**
2. Knowledge about the outputs
3. Knowledge about the inputs



Transfer Learning

Pretraining

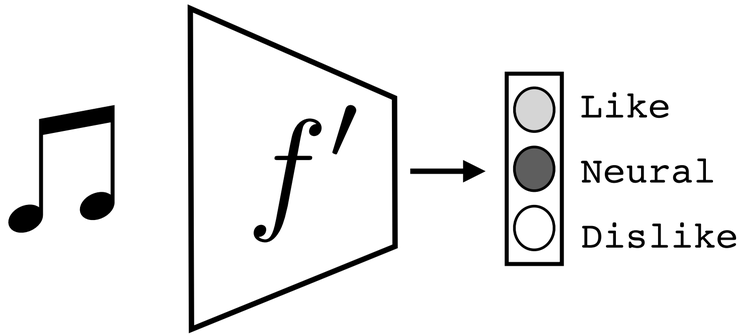
Genre recognition



A lot of data

Adapting

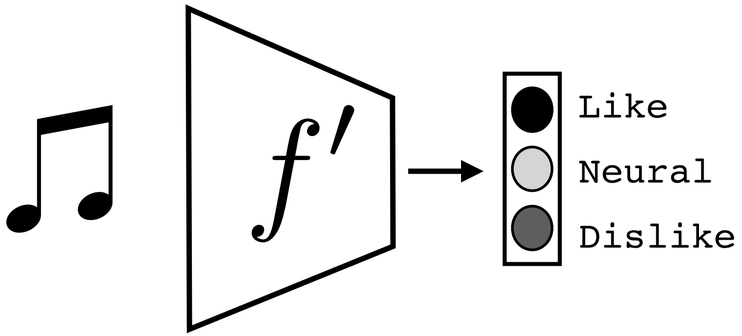
Preference prediction



A little data

Testing

Preference prediction



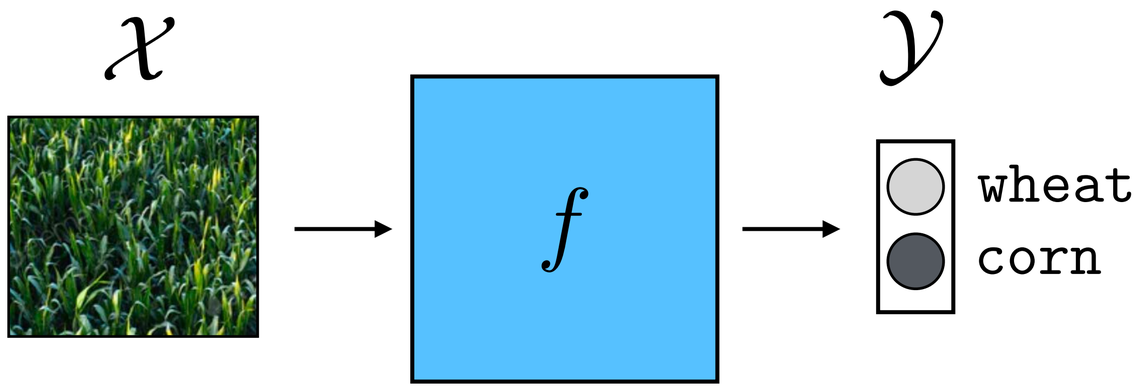
Finetuning

- Pretrain a network on task A, resulting in parameters **W** and **b**
- Initialize a second network with some or all of **W** and **b**
- Train the second network on task B, resulting in parameters **W'** and **b'**

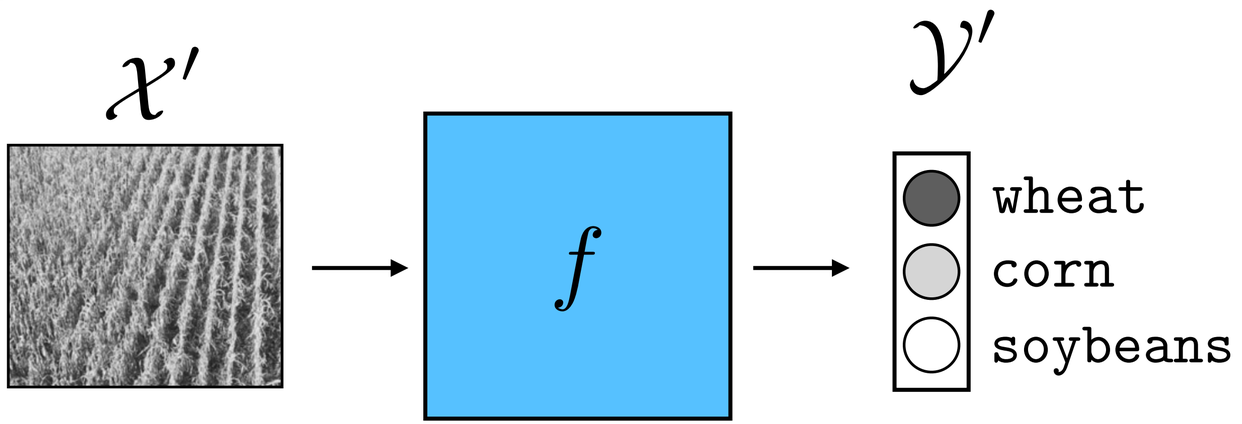
The "learned representation" is the *encoder* (its weights and biases) so that's what we transfer

What if the input/output dimensions don't match?

Pretraining



Finetuning

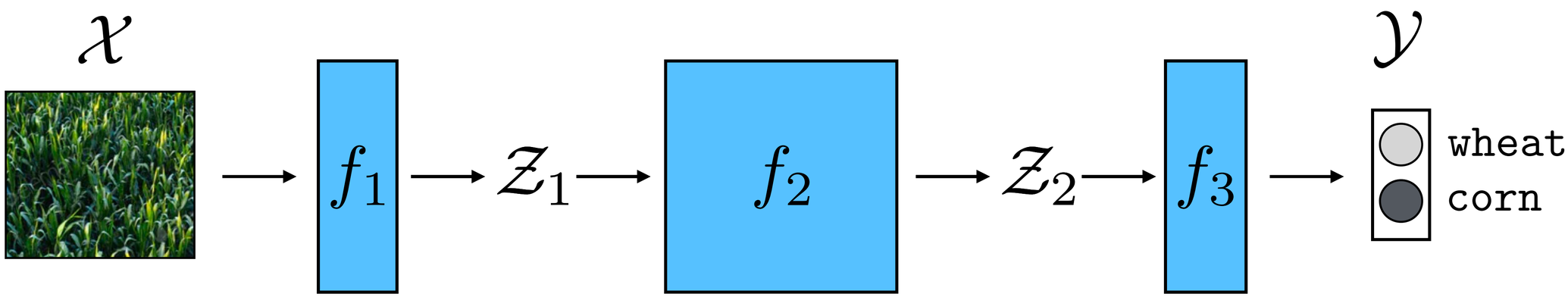


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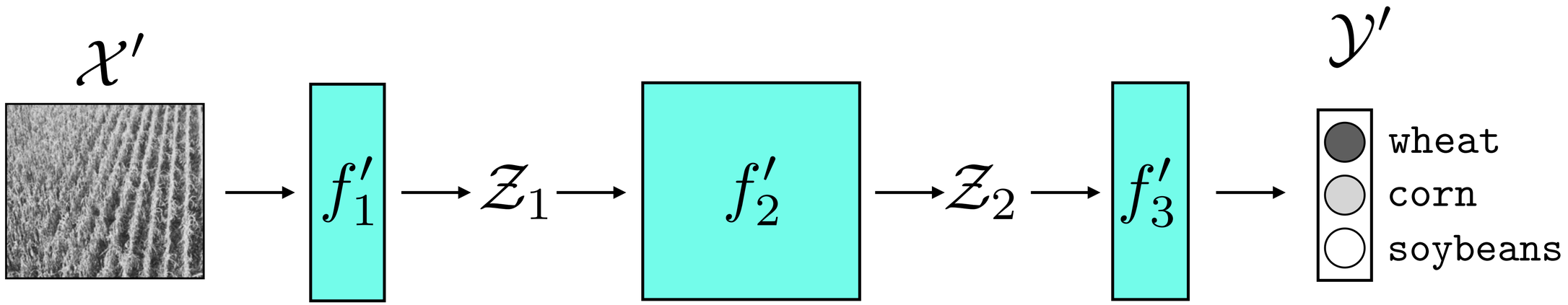
What if the input/output dimensions don't match?

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Pretraining



Finetuning



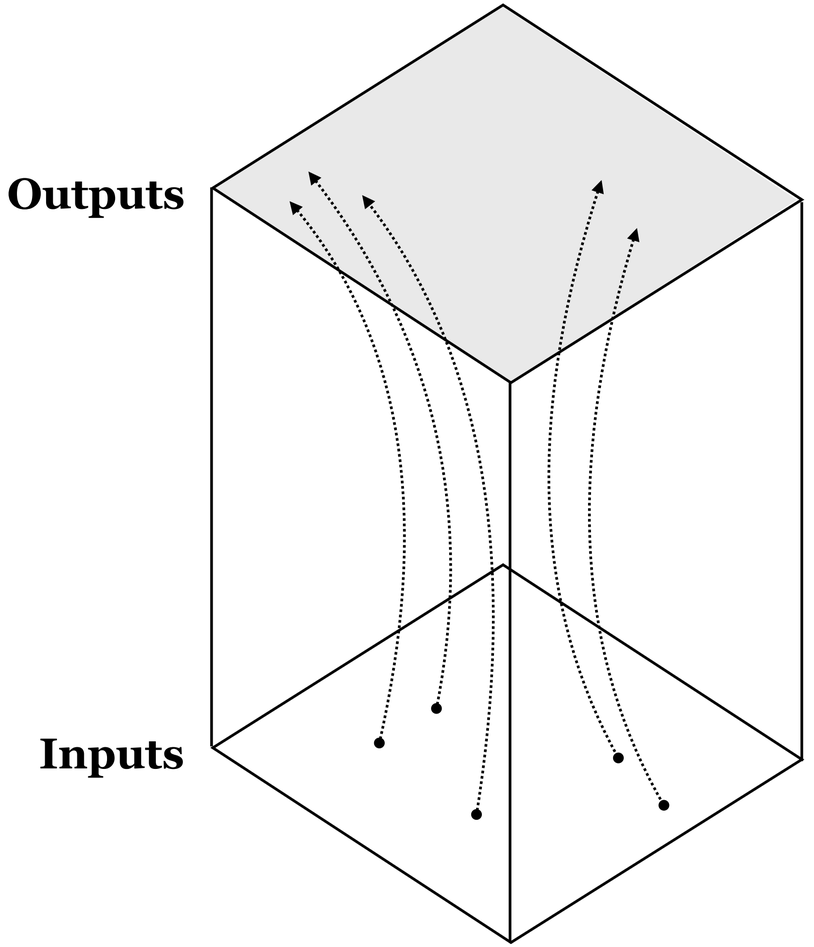
How to avoid forgetting the general representation?

Avoiding overfitting on a small amount of data for a new task while still enabling specialization can be a balancing act. A few methods:

- Freeze most of the network, train only the final layers
- Use a small learning rate
- Early stopping
- Continue to train on the original data as well

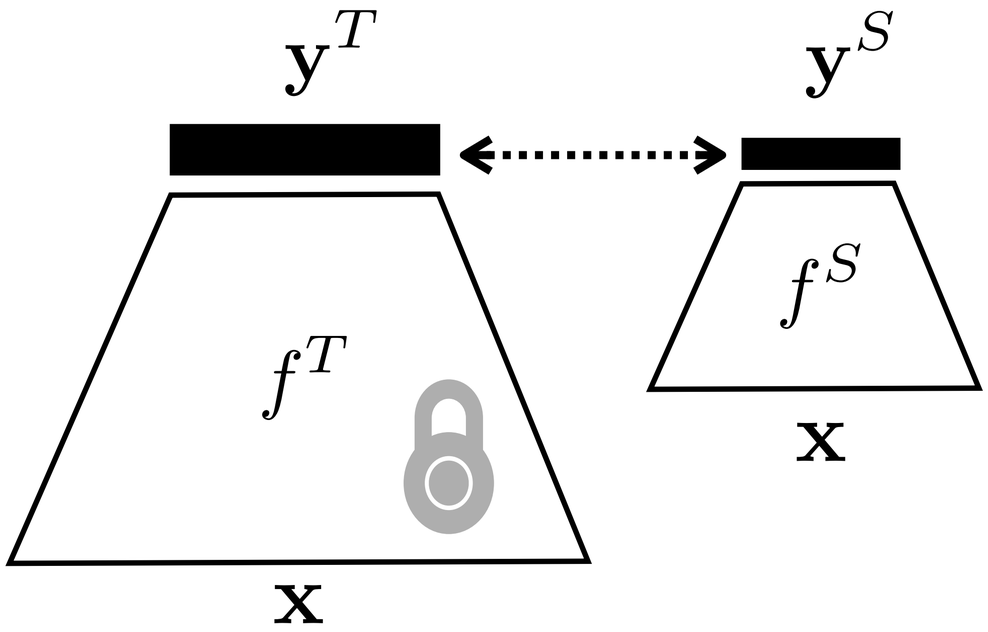
All assume that you have a good, representative validation set for your task (not always possible when you may see distribution shift)

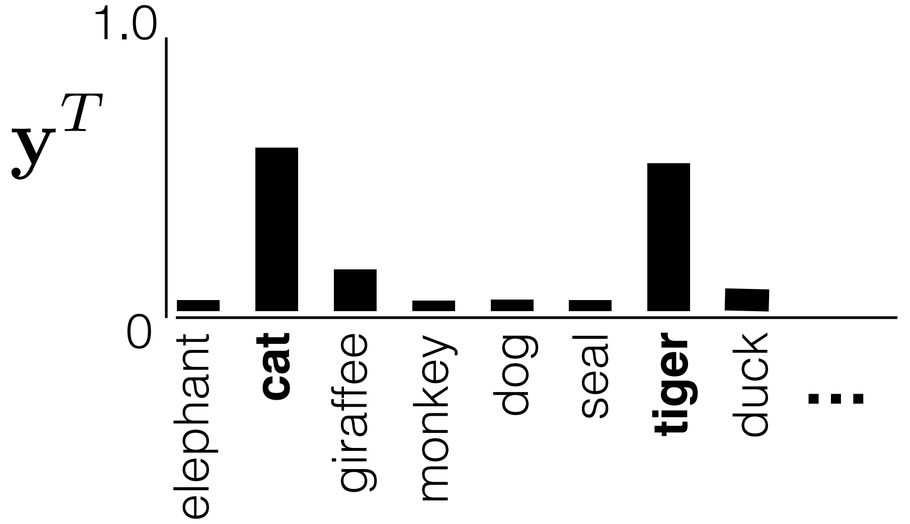
1. Knowledge about the mapping
2. **Knowledge about the outputs**
3. Knowledge about the inputs

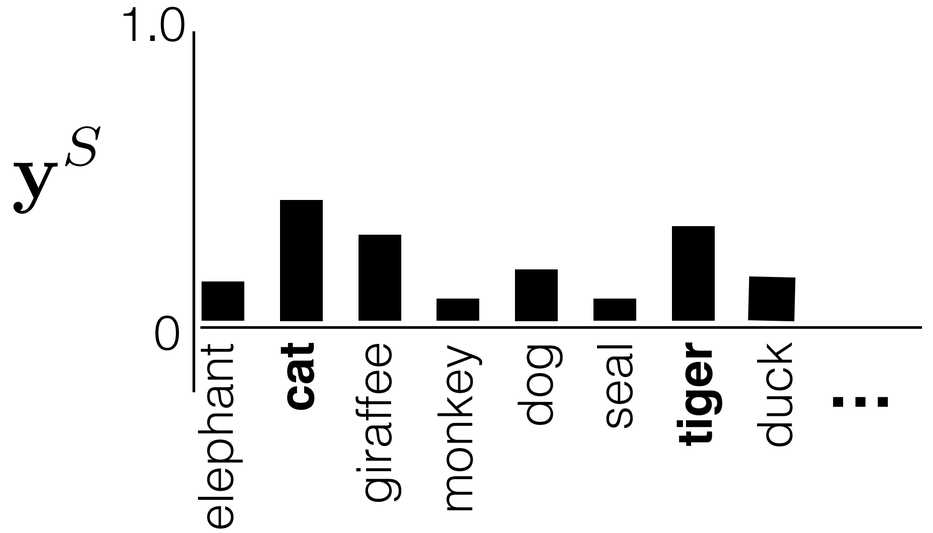


Transfer Learning

Knowledge Distillation [Hinton, Vinyals, Dean, 2015]







$$L(\mathbf{y}^S, \mathbf{y}^T) = -\sum_i \mathbf{y}_i^T \log \mathbf{y}_i^S$$

Knowledge Distillation [Hinton, Vinyals, Dean, 2015]

- The “label” you get for each input is more informative than just the ground truth class
- The “label” (**soft target**) says not just “this is a dog” but “this is a dog that looks a bit like a cat and very different from a bear” (so maybe it’s a small dog).

Hinton paper shows lower train accuracy but higher test accuracy on speech recognition with soft targets

System & training set	Train Frame Accuracy	Test Frame Accuracy
Baseline (100% of training set)	63.4%	58.9%
Baseline (3% of training set)	67.3%	44.5%
Soft Targets (3% of training set)	65.4%	57.0%

Table 5: Soft targets allow a new model to generalize well from only 3% of the training set. The soft targets are obtained by training on the full training set.

Compression vs. Distillation

- Both give us smaller models with similar accuracy to large ones
- Distillation uses SGD to train a smaller model to match characteristics of a larger one
- Compression uses ***pruning*** or ***quantization*** to reduce the computational cost of a large model

Algorithm 1 Prune $k \leq d_{\text{col}}$ weights from row $\mathbf{w}$ with inverse Hessian $\mathbf{H}^{-1} = (2\mathbf{X}\mathbf{X}^\top)^{-1}$ according to OBS in $O(k \cdot d_{\text{col}}^2)$ time.

$M = \{1, \ldots, d_{\text{col}}\}$
for $i = 1, \ldots, k$ **do**
  $p \leftarrow \text{argmin}_{p \in M} \frac{1}{[\mathbf{H}^{-1}]_{pp}} \cdot w_p^2$
  $\mathbf{w} \leftarrow \mathbf{w} - \mathbf{H}^{-1}_{:,p} \frac{1}{[\mathbf{H}^{-1}]_{pp}} \cdot w_p$
  $\mathbf{H}^{-1} \leftarrow \mathbf{H}^{-1} - \frac{1}{[\mathbf{H}^{-1}]_{pp}} \mathbf{H}^{-1}_{:,p} \mathbf{H}^{-1}_{p,:}$
  $M \leftarrow M - \{p\}$
end for

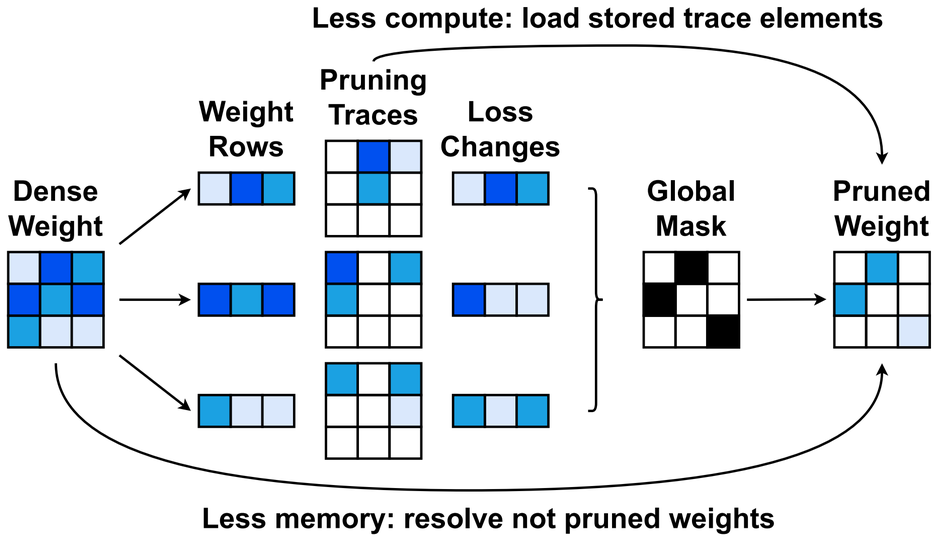


Figure 1: Efficient global OBS using the row-wise results.

Courtesy of Frantar, et al. Used under CC BY.

Optimal Brain Compression: A Framework for Accurate Post-Training Quantization and Pruning, Frantar et al., 2023

Foundation models

[Bommasani et al. 2021] https://arxiv.org/pdf/2108.07258.pdf

"If I have seen further it is by standing on the shoulders of Giants"
— Newton



[*Blind Orion Searching for the Rising Sun* by Nicolas Poussin, 1658]

Image is in the public domain.

AlexNet to AlphaGo Zero: A 300,000x Increase in Compute



The total amount of compute, in petaflop/s-days,[2] used to train selected results that are relatively well known, used a lot of compute for their time, and gave enough information to estimate the compute used.

https://openai.com/blog/ai-and-compute/

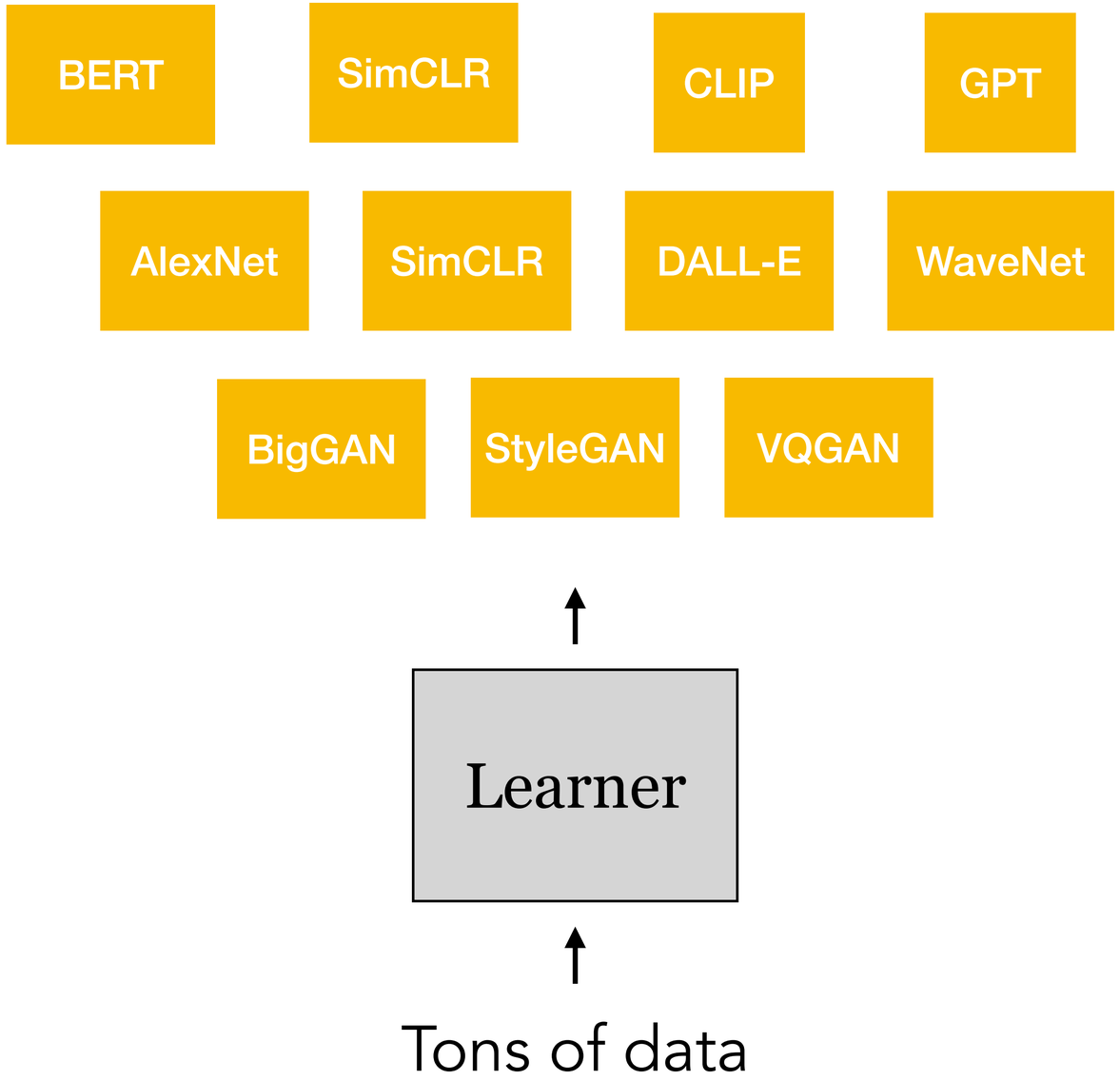
What does it take to train one of these models?

- Dalle-2: **650m training examples** (semi-curated image-text pairs)
- GPT-3: ~**$4.6m to train *once*** (https://lambdalabs.com/blog/demystifying-gpt-3/)
- PaLM: **67 authors**

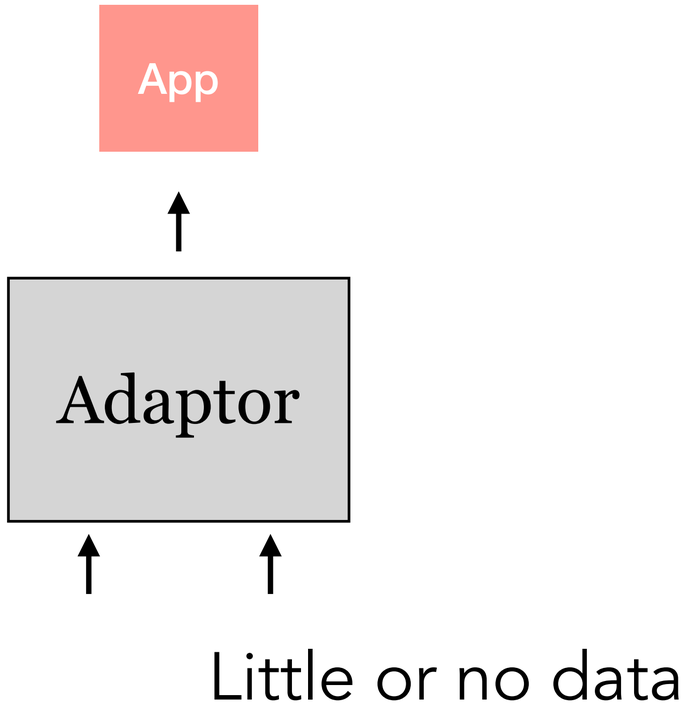
And these numbers are increasing.

But these numbers are small if we think of the result as foundational infrastructure; compare to the cost of building the interstate highway system, or the cost of developing the Mac OS X operating system.

Learn foundation models

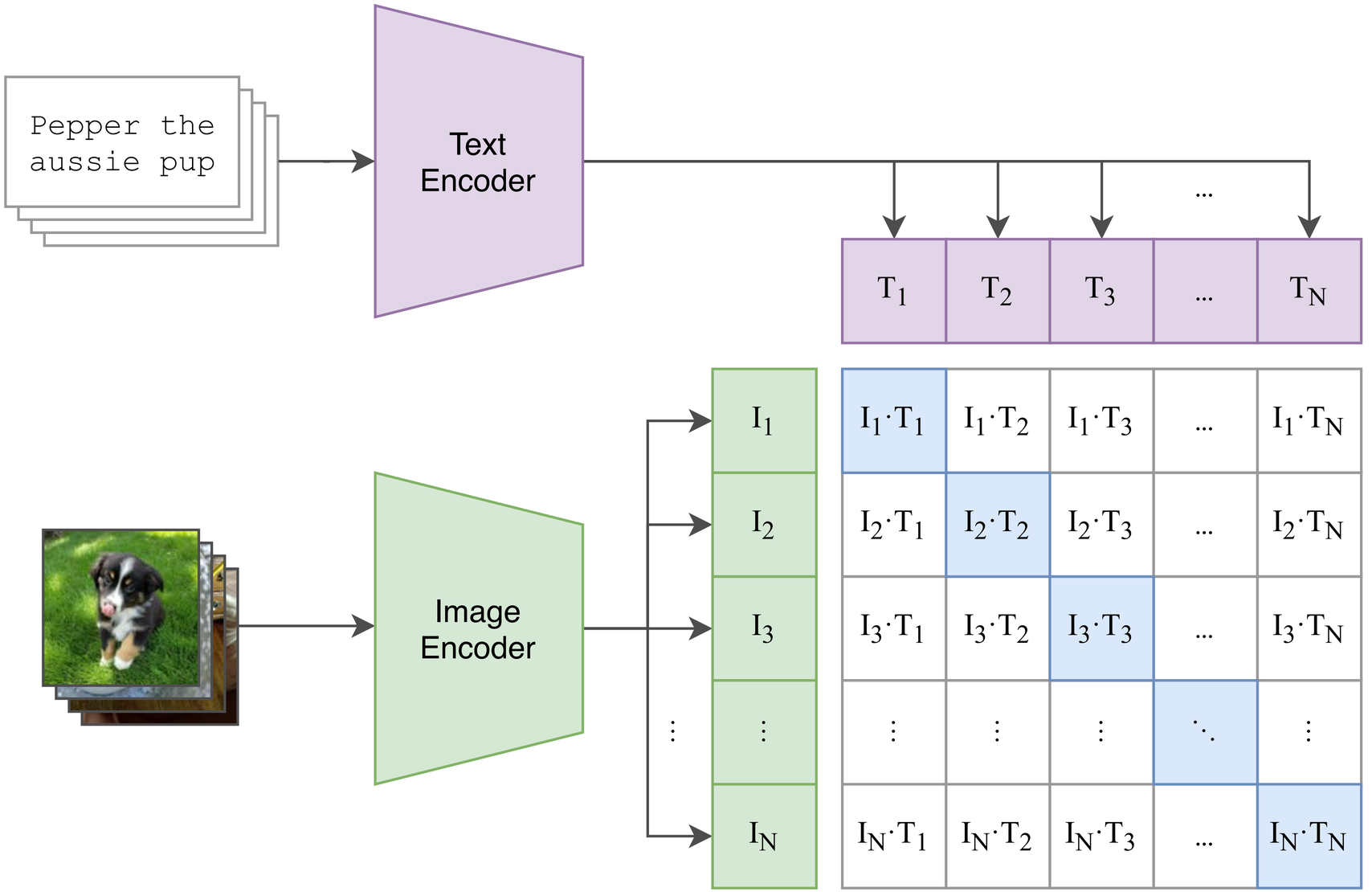


Use/adapt foundations to solve new problems



CLIP [Radford et al., 2021] https://arxiv.org/pdf/2103.00020.pdf

1. **Learner**: Contrastive Pretraining

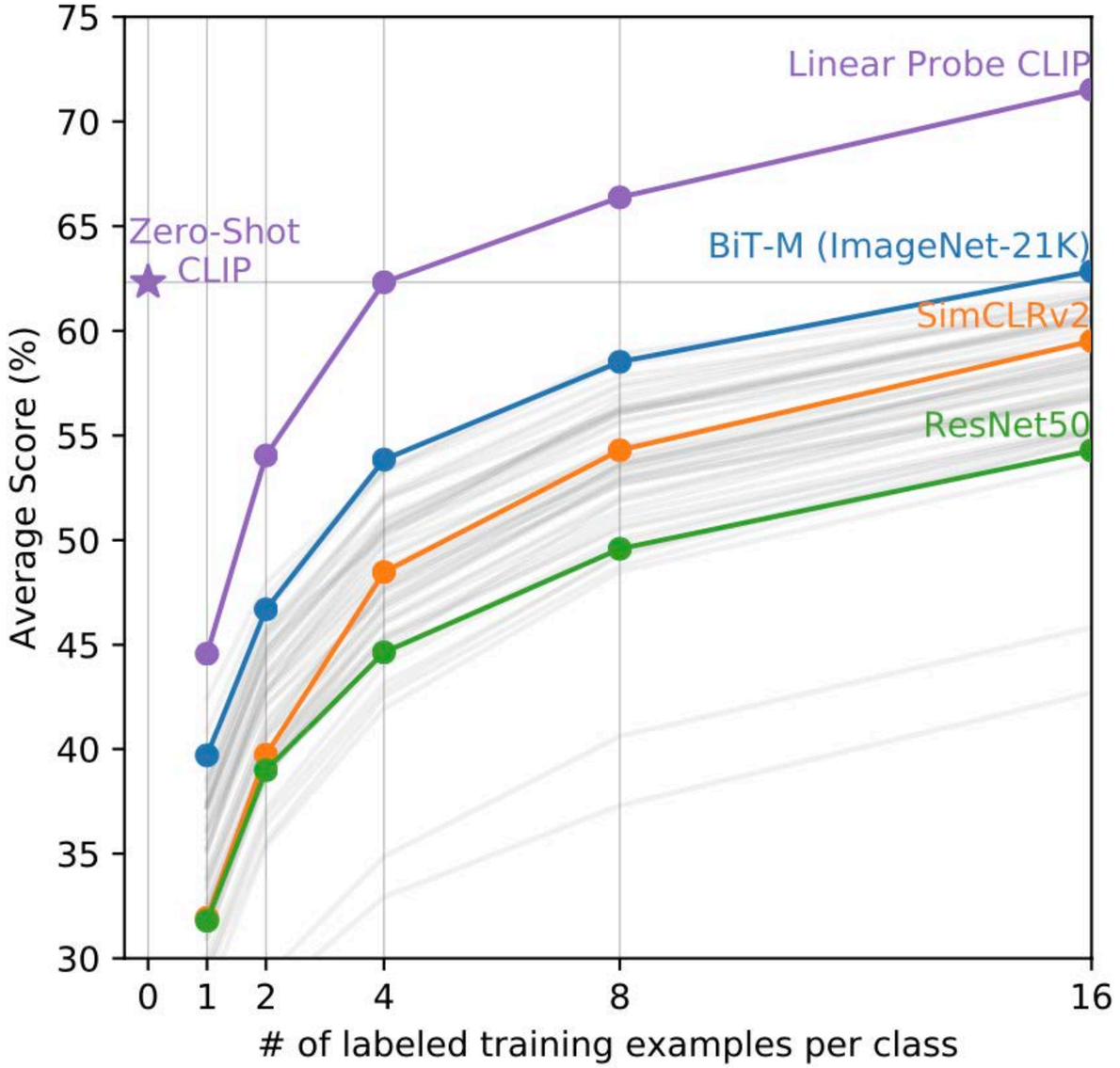


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[https://openai.com/blog/clip/]

CLIP [Radford et al., 2021] https://arxiv.org/pdf/2103.00020.pdf

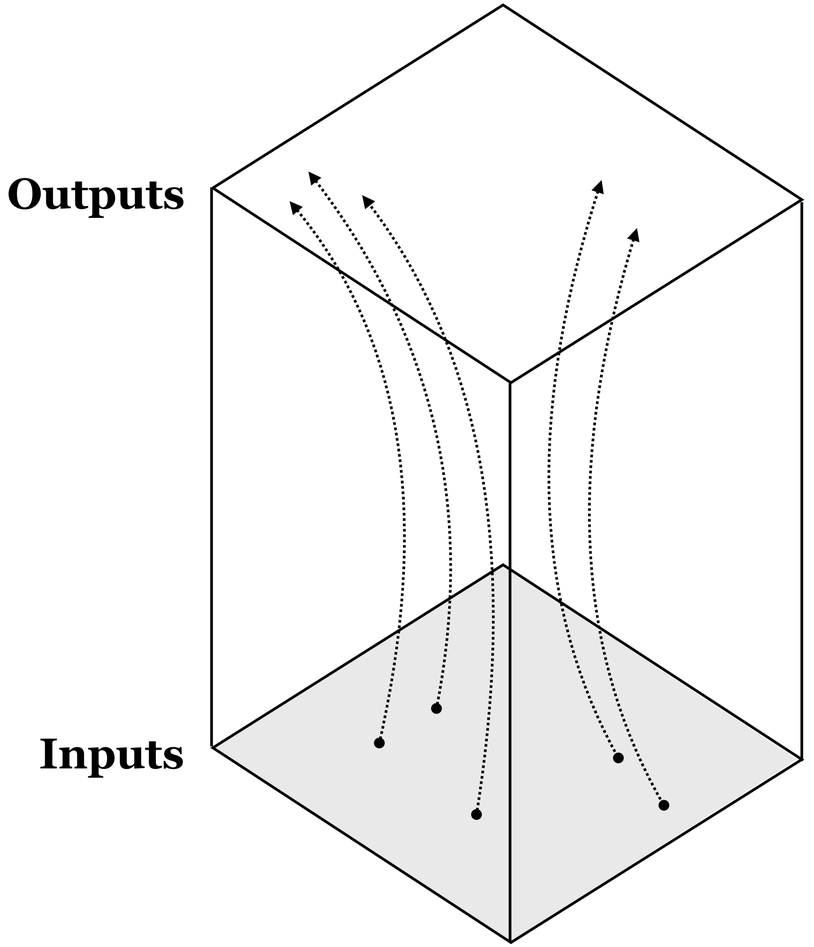
2. **Adaptor**: Linear classifer on top of image encodings



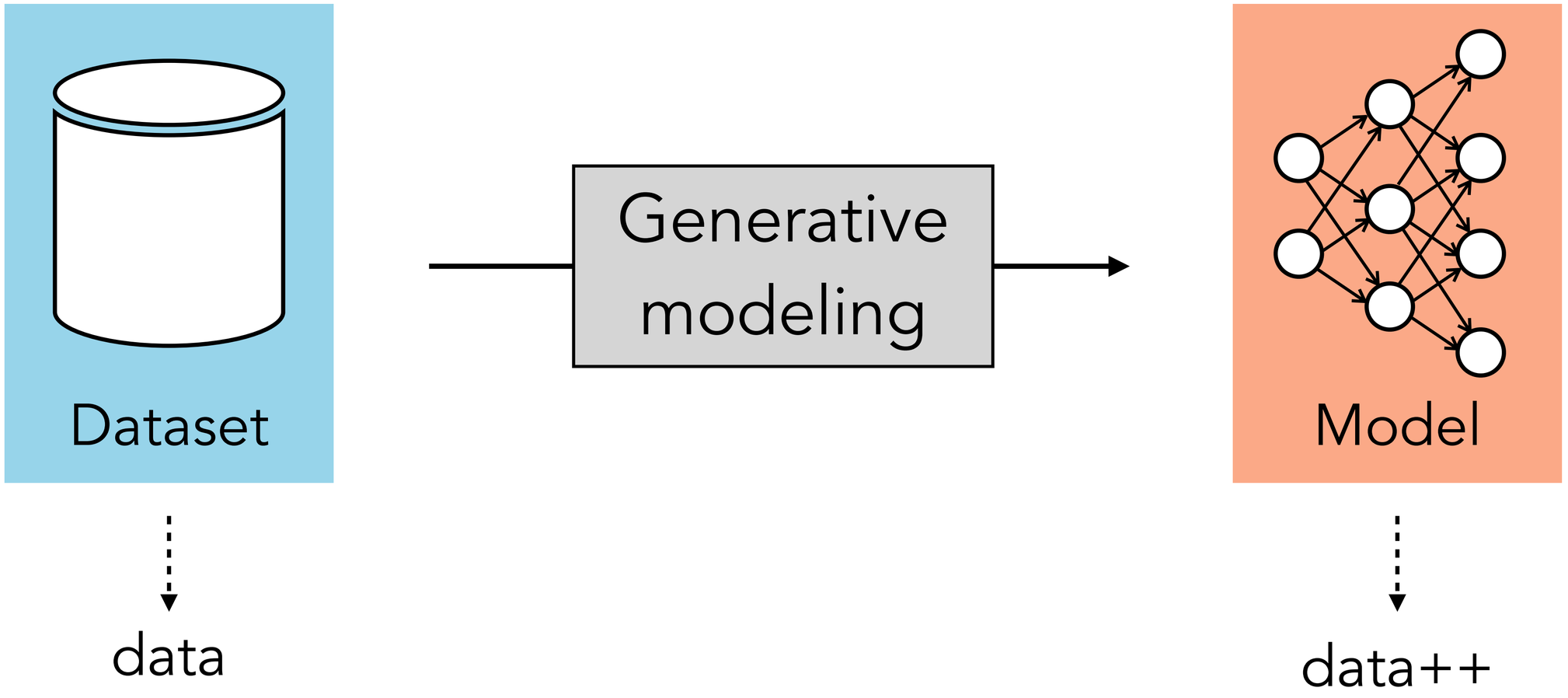
[https://openai.com/blog/clip/]

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1. Knowledge about the mapping
2. Knowledge about the outputs
3. **Knowledge about the inputs**



Transfer Learning



“This release is the culmination of many hours of collective effort to create a single file that compresses the visual information of humanity into a few gigabytes.”

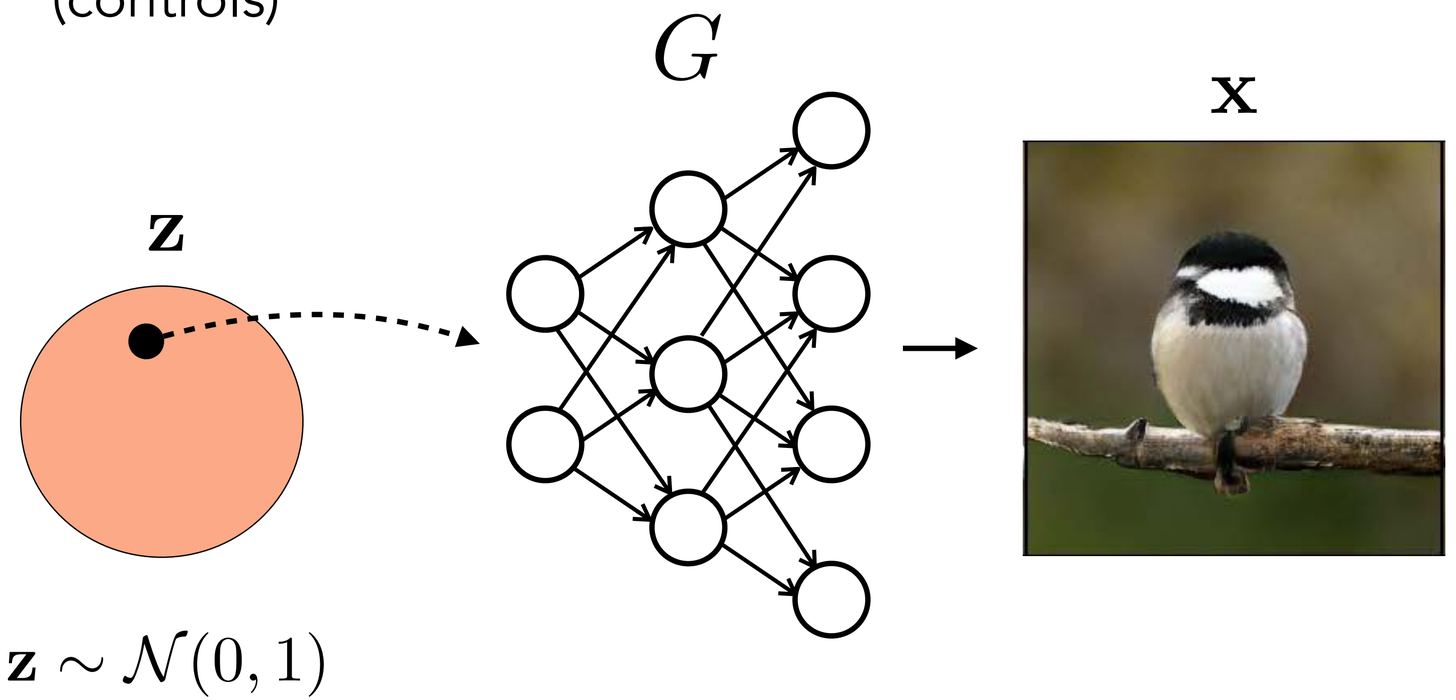
Emad Mostaque [https://stability.ai/blog/stable-diffusion-public-release]

Generative Models

Latent variables

(controls)

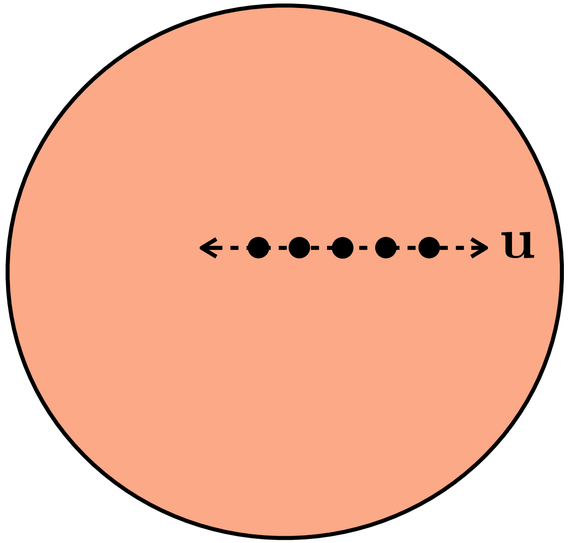
Synthesized Image

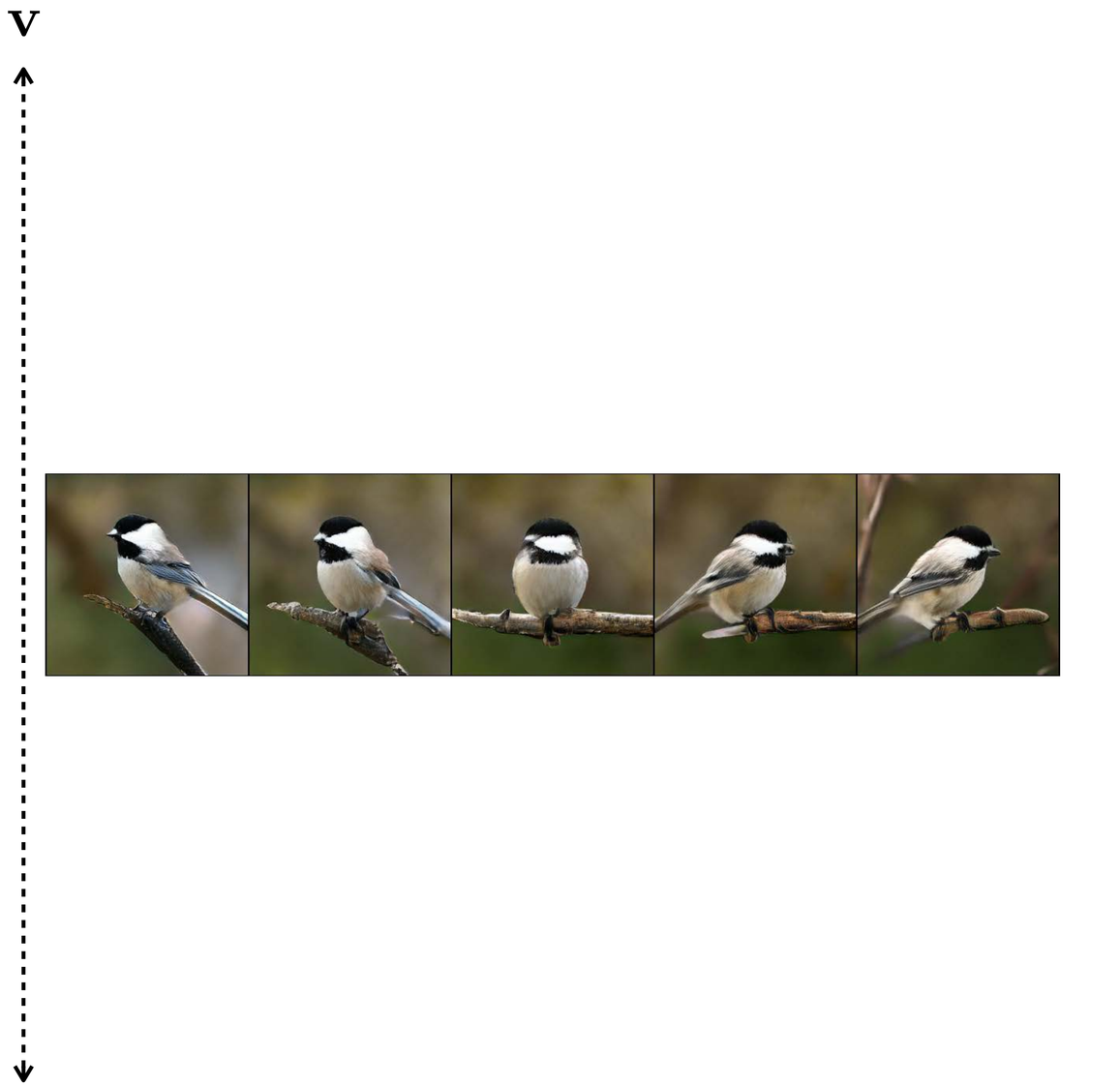


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Synthesized Images

Latent variables
(controls)

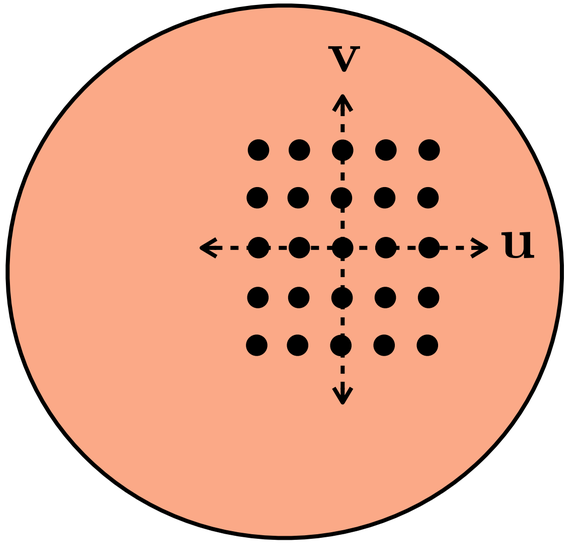


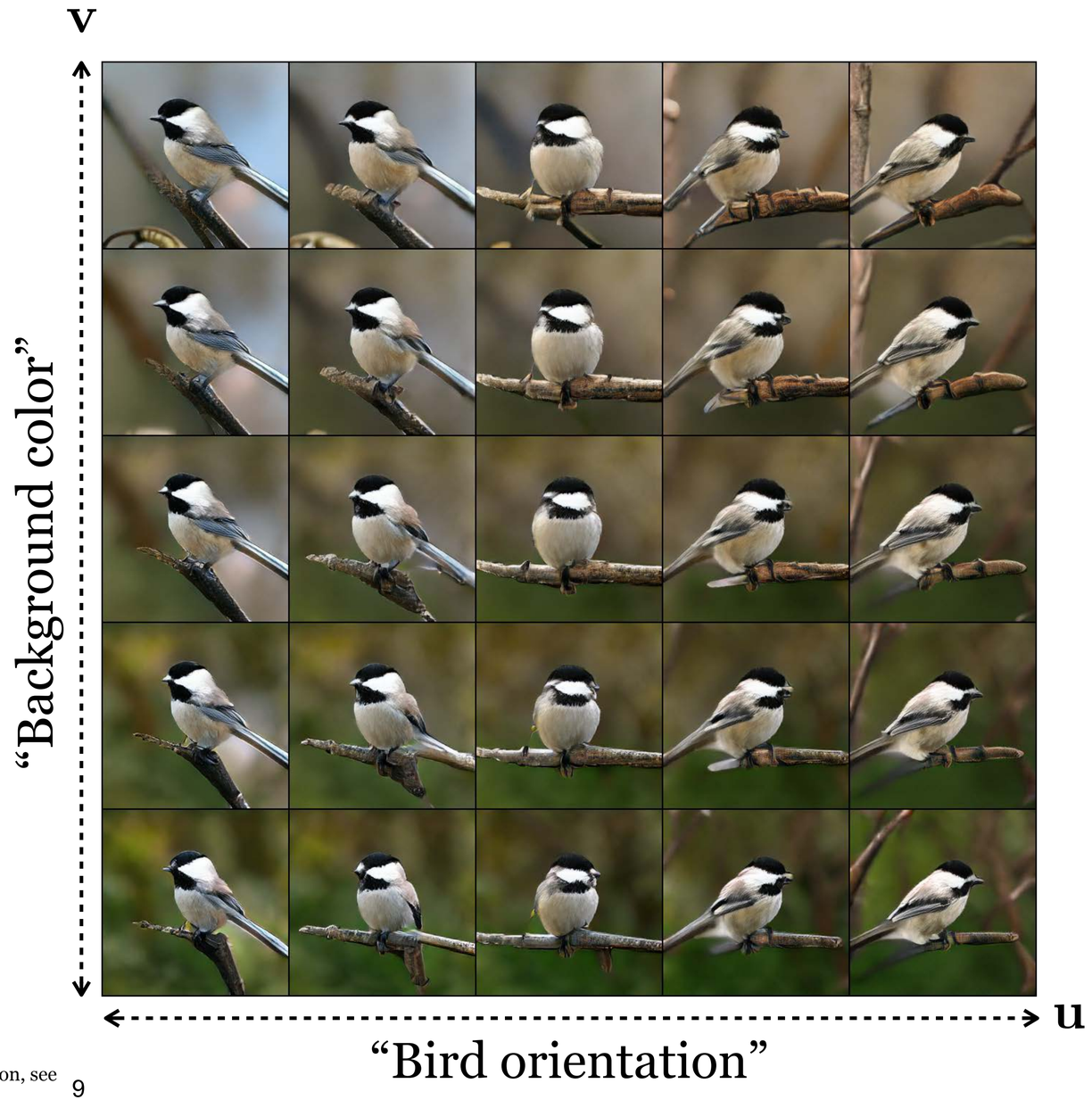


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Synthesized Images

Latent variables
(controls)





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